Entity-Relationship Diagram

travel_itinerary_db

Travel Itinerary and Expense Organizer — MySQL 8.4.13

32 tables · 41 foreign keys · 31 triggers · 4 views · 2 procedures

Legend	
PK / underline	primary key
FK / italic	foreign key
U	unique column
name *	NOT NULL
●—● solid	ON DELETE CASCADE
●--● red	ON DELETE RESTRICT
●- -● amber	ON DELETE SET NULL
●- -● purple	shared primary key (IS-A / weak entity)

Crow's-foot notation	
|| one and only one	|○ zero or one
> one or many	○> zero or many

Full schema — 32 tables, 41 foreign keys

Smart layer: audit + alerts

audit_log (0 rows)

PK	audit_id *	bigint
	table_name *	varchar(40)
	record_id *	varchar(40)
	action *	enum
	field_name	varchar(40)
	old_value	varchar(255)
	new_value	varchar(255)
	changed_by *	varchar(80)
	changed_at *	timestamp

trip_alerts (0 rows)

PK	alert_id *	int
FK	trip_id *	int
	alert_type *	enum
	message *	varchar(200)
	is_read *	tinyint
	created_at *	timestamp

shared key

Hotels module

hotels (8 rows)

PK	hotel_id *	int
	star_rating *	tinyint
	phone	varchar(20)
	email	varchar(100)
	check_in_time *	time
	check_out_time *	time

room_types (16 rows)

PK	room_type_id *	int
FK	hotel_id *	int
	name *	varchar(60)
	bed_type *	enum
	max_occupancy *	int
	base_rate *	decimal(10,2)

(room_type_id, hotel_id)

rooms (32 rows)

PK	room_id *	int
FK	hotel_id *	int
FK	room_type_id *	int
	room_number *	varchar(10)
	floor_no *	int
	status *	enum

RESTRICT

hotel_reservations (8 rows)

PK	reservation_id *	int
FK	trip_id *	int
FK	room_id *	int
FK	booked_by *	int
	check_in *	date
	check_out *	date
	num_guests *	int
	rate_per_night *	decimal(10,2)
	status *	enum
U	confirmation_code	varchar(20)
	special_requests	varchar(255)
	created_at *	timestamp

shared key

hotel_reviews (2 rows)

PK FK	reservation_id *	int
	rating *	tinyint
	title	varchar(100)
	comment	text
	created_at *	timestamp

hotel_payments (10 rows)

PK FK	reservation_id *	int
PK	payment_no *	int
	amount *	decimal(10,2)
	method *	enum
	status *	enum
	transaction_ref	varchar(40)
	paid_at *	timestamp

amenities (8 rows)

PK	amenity_id *	int
U	name *	varchar(40)

hotel_amenities (34 rows)

PK FK	hotel_id *	int
PK FK	amenity_id *	int
	is_complimentary *	tinyint

Core: users, trips, itinerary, money

destinations (7 rows)

PK	destination_id *	int
	name *	varchar(80)
	country *	varchar(80)
	description	text
	best_season	varchar(40)

RESTRICT

saved_destinations (7 rows)

PK FK	user_id *	int
PK FK	destination_id *	int
	saved_at *	timestamp

places (24 rows)

PK	place_id *	int
FK	destination_id *	int
	name *	varchar(100)
	address	varchar(200)
	place_type	varchar(40)

itinerary_days (24 rows)

PK	day_id *	int
FK	trip_id *	int
	day_number *	int
	day_date *	date
	notes	varchar(255)

reviews (8 rows)

PK FK	user_id *	int
PK FK	destination_id *	int
	rating *	tinyint
	comment	text
	created_at *	timestamp

activity_categories (7 rows)

PK	category_id *	int
U	name *	varchar(40)

RESTRICT

SET NULL

itinerary_items (47 rows)

PK	item_id *	int
FK	day_id *	int
	seq_no *	int
FK	category_id *	int
FK	place_id	int
	title *	varchar(100)
	start_time	time
	end_time	time
	estimated_cost *	decimal(10,2)
	status *	enum
	notes	text

SET NULL

bookings (11 rows)

PK	booking_id *	int
FK	item_id *	int
	provider *	varchar(80)
	confirmation_code	varchar(40)
	status *	enum
	amount *	decimal(10,2)

trips (5 rows)

PK	trip_id *	int
FK	owner_id *	int
	title *	varchar(100)
	start_date *	date
	end_date *	date
	budget_total *	decimal(10,2)
	status *	enum
	is_public *	tinyint

RESTRICT

trip_members (15 rows)

PK FK	trip_id *	int
PK FK	user_id *	int
	role *	enum
	joined_at *	timestamp

expense_categories (6 rows)

PK	expense_category_id *	int
U	name *	varchar(40)

RESTRICT

SET NULL

expenses (27 rows)

PK	expense_id *	int
FK	trip_id *	int
FK	item_id	int
FK	expense_category_id *	int
FK	paid_by *	int
	amount *	decimal(10,2)
	expense_date *	date
	description	varchar(200)

users (6 rows)

PK	user_id *	int
U	email *	varchar(100)
	password_hash *	varchar(255)
	full_name *	varchar(80)
	role *	enum
	created_at *	timestamp

RESTRICT

RESTRICT

RESTRICT

RESTRICT

Transport module

carriers (10 rows)

PK	carrier_id *	int
U	name *	varchar(80)
	carrier_type *	enum
	code	varchar(10)

stations (16 rows)

PK	station_id *	int
	code *	varchar(10)
	name *	varchar(100)
	city *	varchar(60)
	station_type *	enum

RESTRICT

RESTRICT

RESTRICT

transport_bookings (14 rows)

PK	transport_id *	int
FK	trip_id *	int
FK	item_id	int
	transport_mode *	enum
FK	carrier_id *	int
FK	booked_by *	int
FK	origin_station_id	int
FK	destination_station_id	int
	depart_at *	datetime
	arrive_at *	datetime
	status *	enum
	fare *	decimal(10,2)
U	confirmation_code	varchar(30)
	created_at *	timestamp

shared key

shared key

shared key

shared key

shared key

flight_details (3 rows)

PK	transport_id *	int
	flight_number *	varchar(10)
	cabin_class *	enum
	terminal	varchar(10)
	baggage_kg *	int

metro_details (2 rows)

PK	transport_id *	int
	line_name *	varchar(40)
	ticket_type *	enum
	num_stops	int

cab_details (5 rows)

PK	transport_id *	int
	vehicle_type *	enum
	driver_name	varchar(60)
	driver_phone	varchar(20)
	vehicle_number	varchar(20)
	pickup_address *	varchar(200)
	drop_address *	varchar(200)
	distance_km	decimal(7,1)
	trip_type *	enum

bus_details (3 rows)

PK	transport_id *	int
	bus_type *	enum
	bus_number	varchar(20)

train_details (1 rows)

PK	transport_id *	int
	train_number *	varchar(10)
	train_name	varchar(60)
	travel_class *	enum
	coach	varchar(10)

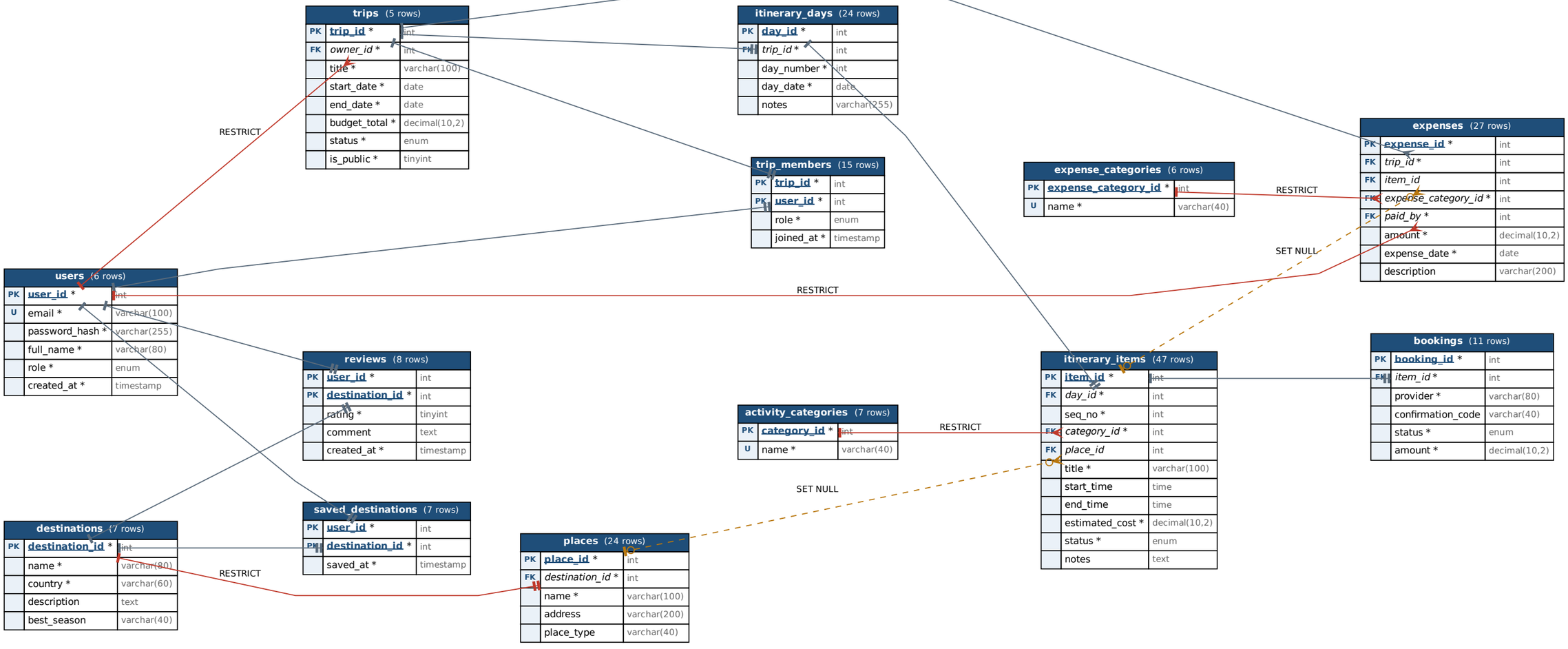
transport_passengers (33 rows)

PK	transport_id *	int
PK	passenger_no *	int
	full_name *	varchar(80)
	age	tinyint unsigned
	seat_no	varchar(10)
FK	user_id	int

SET NULL

RESTRICT

Core: users, trips, itinerary, money — 13 tables



Hotels module — 8 tables

hotels (8 rows)

PK	**hotel_id** *	int
	star_rating *	tinyint
	phone	varchar(20)
	email	varchar(100)
	check_in_time *	time
	check_out_time *	time

room_types (16 rows)

PK	**room_type_id** *	int
FK	*hotel_id* *	int
	name *	varchar(60)
	bed_type *	enum
	max_occupancy *	int
	base_rate *	decimal(10,2)

amenities (8 rows)

PK	**amenity_id** *	int
U	name *	varchar(40)

hotel_amenities (34 rows)

PK	**hotel_id** *	int
PK	**amenity_id** *	int
	is_complimentary *	tinyint

rooms (32 rows)

PK	**room_id** *	int
FK	*hotel_id* *	int
FK	*room_type_id* *	int
	room_number *	varchar(10)
	floor_no *	int
	status *	enum

RESTRICT

(room_type_id, hotel_id)

hotel_reservations (8 rows)

PK	**reservation_id** *	int
FK	*trip_id* *	int
FK	*room_id* *	int
FK	*booked_by* *	int
	check_in *	date
	check_out *	date
	num_guests *	int
	rate_per_night *	decimal(10,2)
	status *	enum
U	confirmation_code	varchar(20)
	special_requests	varchar(255)
	created_at *	timestamp

hotel_payments (10 rows)

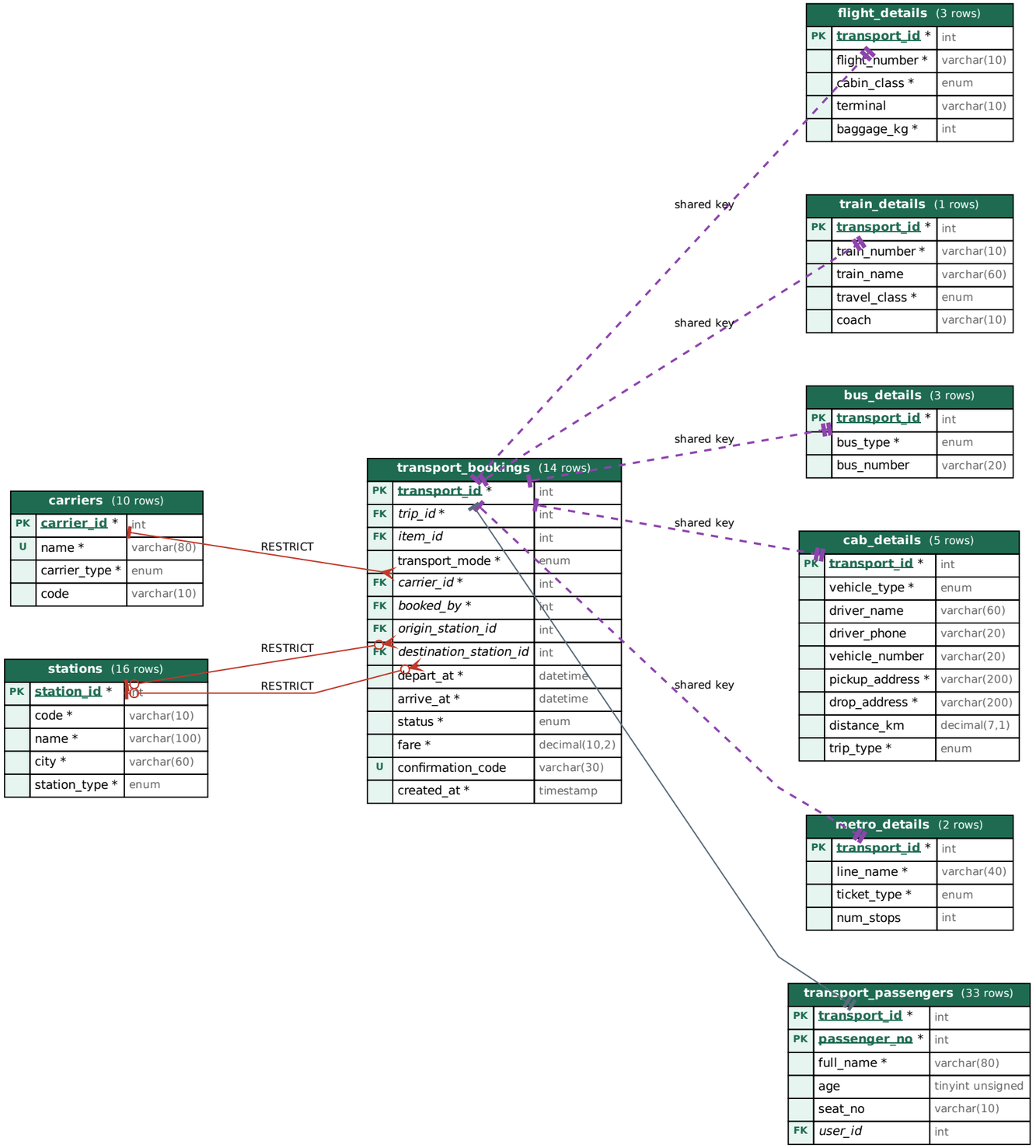
PK	**reservation_id** *	int
PK	**payment_no** *	int
	amount *	decimal(10,2)
	method *	enum
	status *	enum
	transaction_ref	varchar(40)
	paid_at *	timestamp

shared key

hotel_reviews (2 rows)

PK	**reservation_id** *	int
	rating *	tinyint
	title	varchar(100)
	comment	text
	created_at *	timestamp

Transport module — 9 tables



Smart layer: audit + alerts — 2 tables

trip_alerts (0 rows)		
PK	**alert_id** *	int
FK	*trip_id* *	int
	alert_type *	enum
	message *	varchar(200)
	is_read *	tinyint
	created_at *	timestamp

audit_log (0 rows)		
PK	**audit_id** *	bigint
	table_name *	varchar(40)
	record_id *	varchar(40)
	action *	enum
	field_name	varchar(40)
	old_value	varchar(255)
	new_value	varchar(255)
	changed_by *	varchar(80)
	changed_at *	timestamp